

Generation Interconnection System Impact Study Report

For Queue Project Q17-0732

Cauthorn 100 kV
February 02, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0732. The Interconnection Customer proposes a 88.7 MW solar photovoltaic facility interconnecting at the Cauthorn 100 kV substation in the Ohio area. The request was received on January 21, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 341 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,164,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fort Humbug – Little Chute 345 KV	Coronet – Fort Humbug 500 KV	101.4	1,720	5.2
Hilltop – Piney Crk Project 500 KV Line	Caldwell – Piney Crk Project 345 KV	100.4	2,515	11.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Little Chute – Ft Humbug 345 KV	Reconductor line	\$31,245,000	2% (\$646,000)
Hilltop – Piney Crk Project 500 KV Line	Reconductor line	\$180,126,000	2% (\$3,518,000)
	Total allocated to project		\$4,164,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.